

iPSC-Derived Myogenic Progenitor Cells from People with ME/CFS Reveal Altered Genes and Pathways with Drug Repurposing Potential

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Research Article

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Abstract

Background

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a debilitating multisystem disorder characterised by profound fatigue, post-exertional malaise, muscle pain and weakness. Although skeletal muscle abnormalities and mitochondrial alterations have been reported in people with ME/CFS, the underlying biological processes remain poorly understood, and no validated disease models, biomarkers or effective treatments are currently available. This study used patient-derived induced pluripotent stem cell (iPSC) myogenic progenitor (MP) cells to investigate intrinsic transcriptomic abnormalities in ME/CFS and identify potential drug repurposing candidates.

Methods

MP cells were generated from iPSCs derived from people with ME/CFS and healthy controls, and successfully characterised. Global mRNA sequencing was performed to compare transcriptomic profiles between ME/CFS and control MP cells ($n = 6$ ME/CFS; $n = 7$ controls). Differentially expressed genes and pathways were identified, followed by drug repurposing analysis using the Library of Integrated Network-based Cellular Signatures drug database (LINCS2) and supporting literature review.

Results

RNA sequencing identified seven differentially expressed genes at adjusted $p < 0.05$, including *MIR205HG*, *FEZF1*, *HLA-DMB*, *SLC1A2*, *SYT13*, *GALNT4*, and *SLC2A14*. Pathway analysis identified 73 differentially expressed pathways, of which 96% were downregulated in ME/CFS MP cells. Downregulated pathways included those involved in cell cycle regulation, DNA replication, mismatch repair, immune response and muscle cytoskeleton regulation-related pathways. In contrast, upregulated pathways were associated with metabolic reprogramming, including increased reliance on branched-chain amino acids for energy production. These findings suggest intrinsic abnormalities in ME/CFS-derived MP cells, particularly involving metabolic regulation, mitochondrial function and muscle-related cellular processes. Based on the differentially expressed genes, LINCS2-based drug repurposing analysis and literature review identified 22 candidate drugs with potential relevance to ME/CFS. These included safe and widely available agents such as leflunomide, melatonin and midodrine, which target pathways related to antiviral defence, immunomodulation, neurotransmitter modulation, autonomic regulation and vascular function.

Conclusions

Patient-derived iPSC-MP cells provide a useful model for investigating intrinsic skeletal muscle-related abnormalities in ME/CFS. Transcriptomic profiling revealed altered pathways related to cell cycle regulation, immune function, cytoskeletal organisation and energy metabolism, supporting the involvement of mitochondrial and metabolic dysfunction in ME/CFS pathobiology. Drug repurposing analysis identified several clinically relevant candidates that may warrant further functional validation as potential therapeutic options for ME/CFS.

Introduction

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a chronic disease known to affect around 67 million people worldwide [1]. ME/CFS affects the quality of life of both patients and their caregivers, and imposes a significant economic burden estimated to cost up to US\$362 billion annually in the USA [2] and up to \$14 billion in Australia [3]. However, approximately 84–91% of individuals with ME/CFS remain undiagnosed, largely due to the diagnostic complexity

of the condition, heterogeneity, overlap with other disorders, the absence of validated biomarkers and effective treatments that incentivise help seeking [4]. Despite decades of research, there are still no approved drug treatments, biomarkers, or clinical practice guidelines for ME/CFS [5].

Post-exertional malaise (PEM) is recognised as a hallmark of ME/CFS [6]. Muscle pain (myalgia) and weakness, which can worsen for several days following even minor physical exertion, are common and are often reported to be more disabling than the fatigue itself [7]. Reduction of hand grip strength and quadricep strength are frequently observed in ME/CFS patients and correlate with the severity of disease progression [8, 9]. MRI, electron imaging and histology studies have shown damage to skeletal muscle, mitochondrial dysfunction and an increase in anaerobic metabolism in skeletal muscle of ME/CFS patients [6, 10, 11]. While identifying specific molecular targets for ME/CFS remains an ideal goal, it has been largely elusive and target-based drug discovery in ME/CFS has yet to translate into successful clinical treatments [12].

Cells derived from induced pluripotent stem cells (iPSCs) retain the genetic information of the original donor [13], enabling the identification of genetic signatures and pathway alterations that may distinguish patients and healthy cohorts, and which can facilitate the discovery of disease biomarkers and accelerate the development of new therapeutic approaches through drug repurposing and drug discovery. Gene expression signatures (GES) obtained via RNA sequencing from iPSC-derived cells have been successfully used to identify biomarkers and screen potential drug candidates for neurodegenerative diseases such as Alzheimer's disease and frontotemporal dementia [14, 15], as well as major depressive disorder [16, 17], for which primary cells are difficult or impossible to obtain. One of the key advantages of using iPSC-derived cells is that transcriptomic data from RNA-sequencing of these cells can allow us to bypass the lack of known pathophysiological mechanisms to identify potential novel therapeutics [18]. When combined with drug databases, transcriptomic profiles enable the screening of drugs that may reverse the gene expression differences observed between cells derived from patients and healthy controls. Drug repurposing candidates include those that are generally already approved or undergoing clinical trials, with established mechanisms of action and known safety profiles. This approach significantly shortens the timeline for therapeutic development, which is typically lengthy and uncertain in conventional drug discovery. Thus, iPSC-derived cells from ME/CFS patients and healthy donors can serve as a powerful tool for uncovering disease-specific gene expression signatures in muscle cells and identifying targeted drug candidates capable of reversing transcriptomic abnormalities in muscle cells of people with ME/CFS.

The present study is the first to use induced pluripotent stem cell - derived myogenic progenitors (iPSC-MPs) generated from blood samples collected from both people with ME/CFS and healthy controls. The aim is to discover gene and pathway signatures associated with the disease, and to identify transcriptomic targeted drug candidates for repurposing based on the alignment of data from transcriptomic signatures, the LINCS2 drug database, and literature review. Findings from this study will contribute to advancing drug repurposing efforts, bolster the translational potential of identified approved drugs, and add to the development of diagnostic biomarkers and novel therapeutic strategies for ME/CFS.

Methods details

This study received ethics approval from Barwon Health Human Research Ethics Committee (20/202) and Deakin University Human Research Ethics Committee (2021 – 165), Australia.

Study design, participant recruitment and sample collection

A case-control study was conducted at the School of Medicine, Deakin University, Australia. Thirteen participants, including six ME/CFS cases and seven healthy controls were recruited to donate blood samples for generating iPSCs. Potential participants with ME/CFS were identified through the Geelong ME/CFS Support Group and Emerge Australia. Control participants were recruited from the community via advertisement flyers. After a telephone screen, eligible participants were invited to join the study. Recruitment occurred between June 2021 and December 2023. Inclusion criteria included: aged 18 to 65 years, diagnosed with ME/CFS by an independent physician and fulfil the criteria for ME/CFS using the

Canadian Consensus Criteria [19], and have the capacity to consent to and comply with study procedures. Participants currently under any form of therapy for ME/CFS needed to have been on that therapy for at least 4 weeks prior to enrolment. Exclusion criteria included known or suspected active and unstable systemic medical disorder, acute suicidality, current diagnosis of a psychotic disorder (including bipolar disorder, substance abuse/dependence, eating disorder, or personality disorder), or a diagnosis of epilepsy. Trained researchers conducted the screening assessments.

Peripheral blood samples were collected from all participants by a trained nurse, processed to isolate peripheral blood mononuclear cells (PBMCs), and subsequently reprogrammed into iPSCs. The pluripotency of the iPSCs was validated using standard quality control assays. These cells were then differentiated into myogenic progenitor cells (MPs) and used for RNA sequencing, gene signature analysis, and *in silico* drug screening (Fig. 1).

Hand Grip strength test

Repeat hand grip strength was tested using a hand-held dynamometer. Participants were seated comfortably with their forearms resting on the arms of the chair and asked to squeeze the dynamometer for as long and as tightly as possible, then rest for 30 seconds and repeat another two times. The process was then repeated for the other hand.

Karnofsky Performance Status Measurement

The functional status of participants was assessed using the Australian-modified Karnofsky Performance Status (AKPS) scale, a validated instrument for quantifying a patient's ability to carry out daily activities and self-care functions [20]. The AKPS score ranges from 100 (normal, no complaints or evidence of disease) to 0 (death), in 10-point increments reflecting progressive levels of disability. Participants were asked to describe their ability to perform routine work and personal activities, along with any limitations related to fatigue, pain, or other disabling symptoms. Scoring adhered to the original Karnofsky criteria, with adjustments where appropriate to reflect chronic illness populations, which were used to assess overall functional capacity. Lower scores among ME/CFS participants indicated greater disability, consistent with disease severity, whereas healthy controls were expected to score within the 80–100 range.

Multidimensional Fatigue Scale assessment

Fatigue severity and its impact on daily functioning were assessed using the Multidimensional Fatigue Scale (MFI), a validated instrument commonly used in chronic illness populations, including ME/CFS [21]. The MFI is a 20-item self-report questionnaire that evaluates the perceived impact of fatigue on five key domains: general fatigue, physical fatigue, reduced motivation, reduced activity, and mental fatigue. Respondents use a scale ranging from 1 to 7 to indicate how much they agree or disagree with certain statements regarding fatigue to represent their experiences. Scores were summed up to produce a total fatigue impact score (range: 0–84), with higher scores indicating greater fatigue-related impairment. The MFI assessment was administered either electronically or on paper, depending on participant preference and accessibility.

PBMC isolation from participant blood samples and generation of iPSCs

Peripheral blood samples were collected from participants using EDTA vacutainer tubes (Becton Dickinson Sciences). Samples were diluted 1:2 (v/v) with phosphate-buffered saline (PBS, Gibco) containing 2% fetal bovine serum (FBS, Gibco) and 2 mM EDTA (Invitrogen), then transferred to SepMate™-15 tubes (STEMCELL Technologies) preloaded with Lymphoprep (STEMCELL Technologies). Following centrifugation at $1,200 \times g$ for 10 min at room temperature, the PBMC layer was collected and transferred to a fresh tube. PBMCs were washed once with wash buffer and centrifuged at $300 \times g$ for 10 min. To remove residual erythrocytes, the cell pellet was briefly resuspended in ice-cold Milli-Q water, followed by an additional wash in wash buffer and centrifugation at $300 \times g$ for 10 min. The resulting PBMC pellet was cultured in

StemSpan™ SFEM II medium (STEMCELL Technologies) supplemented with StemSpan™ erythroid expansion supplement (STEMCELL Technologies). Reprogramming was performed using the Epi5™ Episomal iPSC Reprogramming Kit (Thermo Fisher Scientific) delivered via the Neon™ Transfection System (Thermo Fisher Scientific) according to the manufacturer's instructions. Transduced cells were plated onto Geltrex™-coated (Gibco) culture dishes and maintained in complete mTeSR™ Plus medium (STEMCELL Technologies) with daily medium changes. Individual iPSC colonies with characteristic morphology were manually selected and expanded in Geltrex-coated flasks in complete mTeSR™ Plus medium. Passaging was performed using ReLeSR™ (STEMCELL Technologies). iPSCs at passages 5 and 10 were cryopreserved in liquid nitrogen for long-term storage. Characterisation of all iPSC lines was conducted between passages 10 and 12 to confirm pluripotency and genomic stability. Cell stocks were stored in liquid nitrogen for subsequent experiments.

Karyotyping

Cells at passages 10–12 were dissociated using TrypLE (Gibco), centrifuged at 300 x g for 5 min, and cell pellet was stored at -20°C until analysis. Karyotyping analysis was performed using the Illumina Infinium GSA-24 v3.0 array and aligned to the GRCh38/hg38 human reference genome (Dec 2013) (Victorian Clinical Genetics Services, VIC, Australia).

Mycoplasma testing

Mycoplasma contamination was assessed using the MycoAlert® Mycoplasma Detection Kit (Lonza Bioscience) following the manufacturer's protocol. Cells were cultured in T25 flasks containing 8 mL of medium without a medium change for 48 h. Subsequently, 1 mL of conditioned medium was collected for testing. Results were interpreted against assay negative and positive controls, with ratios < 0.9 indicating a negative result and > 1.2 indicating a positive result (Supplementary Table 1).

Flow cytometry analysis

Cells at passages 10–12 were harvested using TrypLE and stained with BD Horizon™ Fixable Viability Stain 570 (Becton Dickinson Sciences) for 15 min to assess viability. Then, cells were fixed with 4% paraformaldehyde (Thermo Fisher Scientific) for 15 min, followed by incubation with conjugated antibodies against CD9, SSEA4, and EPCAM for 30 min. Samples were analysed on a BD FACSCanto™ (Becton Dickinson Sciences).

Pluripotency marker immunocytochemistry

Cells were seeded into Geltrex-coated 24-well plates and grown to ~ 60% confluency. After washing with warm PBS, cells were fixed in 4% paraformaldehyde (ProSciTech) for 15 min at room temperature, permeabilised with 0.5% Triton X-100 (Sigma-Aldrich) in PBS for 15 min and blocked with blocking buffer (3% bovine serum albumin (Sigma-Aldrich) in PBS) for 1 h. Primary antibodies diluted in blocking solution were applied for 3 h at room temperature. Following three washes in PBS, cells were incubated with secondary antibodies diluted in blocking buffer for 1 h, then washed a further three times with PBS. Nuclei were counterstained with NucBlue® (Hoechst 33342; Invitrogen) during the final wash for 5 min. Imaging was performed using an EVOS M700 Imaging System (Thermo Fisher Scientific) with a 10x objective; images are unprocessed, with scale bars representing 275µm.

Germ layer differentiation

Trilineage differentiation of iPSCs into ectoderm, mesoderm, and endoderm was performed using the STEMdiff™ Trilineage Differentiation Kit (STEMCELL Technologies). iPSCs were seeded on Matrigel-coated (Corning) plates in mesoderm/endoderm plating medium or ectoderm medium. After 24 h, lineage-specific STEMdiff™ medium was added, with daily medium changes until day 5 for mesoderm and endoderm or day 7 for ectoderm. Cells were passaged on day 5 (mesoderm and endoderm) or day 7 (ectoderm) and cultured for a further 2 days. Cells were then fixed with 4%

paraformaldehyde in PBS for immunofluorescence analysis or harvested in RLT buffer (Qiagen) for gene expression. Imaging was performed on an EVOS M700 Imaging System (Thermo Fisher Scientific,) using a 20× objective; images are unprocessed, with scale bars representing 150 µm.

Proliferation

Cells were seeded into 24-well plates coated with Geltrex and cultured for 72 hours in complete mTeSR™ Plus medium, with daily medium changes. Cells were harvested in QuickExtract DNA Extraction Solution (Biosearch Technologies), and gDNA was extracted according to the manufacturer's instructions. Total DNA was quantified using a NanoDrop spectrophotometer (Thermo Fisher Scientific) at 0, 24, 48, and 72 hours. Analysis shows normalised values to 0h (total number of cells seeded = amount of DNA seeded in each well).

RNA Extraction and quantitative PCR

RNA was extracted using RNeasy® mini kits (Qiagen) and reverse-transcribed to produce cDNA using Maxima H Minus first strand cDNA synthesis kit (Thermo Fisher Scientific) following the manufacturer's instructions. The Quant-iT™ OliGreen® ssDNA Assay Kit (Invitrogen) was used to quantify the cDNA concentration in each sample as per the manufacturer's instructions. Real-time quantitative PCR (qPCR) was used to measure the expression of specific genes, as shown in Supplementary Table 2. The experiments were performed on a QuantStudio 3 Real-time PCR system (Thermo Fisher Scientific), using the following protocol: 95° for 7 min, 40 cycles of 95° for 30s and 60° for 1 min followed by data acquisition, 60° for 30s, 55–95°, data acquisition and 20° for 10s. Gene expression data was quantified using the $\Delta\Delta C_t$ method normalised to the measured cDNA concentration and housekeeping gene GAPDH expression levels, then categorised into ME/CFS and control groups.

MP differentiation

Myogenic progenitor (MP) cells were differentiated from iPSCs. Briefly, iPSCs were seeded at a density of 3.4×10^6 cells per well in Matrigel-coated 6-well plates using DMEM/F-12 GlutaMAX (Gibco) supplemented with STEMdiff™ Myogenic Progenitor supplements A, B, C or D (STEMCELL Technologies). Media was changed daily as follows: Medium A (days 0–1), Medium B (days 2–3), Medium C (days 4–5), and Medium D (days 6–29). On day 29, MPs at passage 0 (P0) were passaged into P1 using TrypLE and collagenase (STEMCELL Technologies), then seeded at 1.92×10^5 cells per well in Matrigel-coated 6-well plates and cultured in DMEM (1000 mg/L D-Glucose; STEMCELL Technologies) supplemented with 10% MyoCult™-SF Expansion 10X Supplement (STEMCELL Technologies). Y-27632 (10 µM; ROCK inhibitor, STEMCELL Technologies) was included only in the seeding medium immediately after passaging and was omitted from all subsequent medium changes. MPs at P1 were subsequently passaged to P2 using Accutase (Gibco), seeded at 9.6×10^4 cells per well in Matrigel-coated 6-well plates, and cultured in DMEM (1000 mg/L D-Glucose) supplemented with 10% MyoCult™-SF Expansion 10X Supplement. Cells were cryopreserved in CryoStor CS10 (STEMCELL Technologies) at P2 and stored in liquid nitrogen for downstream applications. Immunocytochemistry was performed on P2 MPs to characterise the progenitor markers MyoD and MyoG (as previously described).

RNA sequencing

Total RNA from iPSCs at passage 10–12 and MPs at passage 2 were harvested using TRIzol (Life Technologies) with Qiagen RNeasy column purification (Qiagen), following the manufacturers' instructions. RNA-seq libraries were prepared using the Illumina Total RNA with Ribo-Zero Plus Library Prep Kit according to the manufacturer's instructions. The sequencing was performed on the Illumina NovaSeq X platform, generating 150 bp paired-end reads with a depth of 100 million reads per sample for whole transcriptome analysis.

Differential gene expression analysis

Raw sequencing data in FASTQ format were processed using the Galaxy Docker Image locally [22]. Quality control and adapter trimming of raw reads were performed using fastp [23]. Filtered reads were aligned to the human reference

genome (GRCh38) using HISAT2 [24] in RF-stranded mode. Gene-level quantification was performed using featureCounts [25]. Gene expression counts were compiled into an $m \times n$ matrix, where m represents genes and n represents samples ($n = 13$), for downstream differential expression analysis. Normalisation was performed using the median-of-ratios method implemented in the DESeq2 package in R [26]. Genes with low expression levels-defined as < 0.5 count per million (CPM) in more than 30% of samples-were excluded. Differential expression analysis was conducted using DESeq2, with significance determined by an adjusted p-value (false discovery rate, FDR) < 0.05 , using the Benjamini-Hochberg correction for multiple testing. To account for unmeasured confounding variables, surrogate variable analysis was conducted using the sva package in R [27], applying the model formula: “~ Diagnosis + Sample Type”. Biological function information of differentially expressed genes was extracted from GeneCards® - The Human Gene Database (www.genecards.org). Raw sequencing data in FASTQ format were processed using the Galaxy Docker Image locally [22].

Pathway enrichment analysis

To identify biologically relevant pathways enriched among the differentially expressed genes, gene set enrichment analysis (GSEA) was performed using the clusterProfiler package in R. KEGG pathway annotations were used, and pathways with an adjusted p-value < 0.05 were considered significantly enriched. Gene Ontology (GO) annotations were also used to analyse the top pathways and their distribution across different biological categories. Visualisation of enriched pathways was conducted using the Cytoscape platform [28]. Nodes represented individual pathways, and a q-value threshold of 0.05 was applied to highlight the most significantly enriched pathways.

In silico identification of drug repurposing candidates

In silico drug repurposing was conducted using a subset of the top 100 up- and down-regulated genes based on false discovery rate (FDR). These genes were queried against the Library of Integrated Network-based Cellular Signatures (LINCS2) database using the signatureSearch [29] package in R. Compounds were ranked based on the normalised connectivity score (NCS), with lower NCS values indicating stronger inverse associations with the gene expression signature, suggesting potential to reverse disease-associated gene expression patterns. Compounds with a weighted connectivity score (WTCS) FDR < 0.05 were further investigated.

To prioritise drug candidates for repurposing, a literature review was performed on the top-ranking compounds. Exclusion criteria included investigational drugs, agents with black box warnings, veterinary medications, topical treatments, and cytotoxic chemotherapy agents. This filtering ensured that shortlisted drugs had the highest translational potential for safe repurposing in the targeted disease context.

Statistical analysis

Differential gene expression, pathway enrichment and drug *in silico* identification analyses were described above. qPCR results were analysed using the $\Delta\Delta C_t$ method and normalised to cDNA concentration, then categorised into ME/CFS and control groups. Participants' clinical parameters, gene expression levels, and proliferation assay results were recorded and grouped accordingly. Multiple t-test was used to assess data normality and Mann-Whitney u-test was applied to non-normally distributed data statistical comparisons. These analyses were performed using GraphPad Prism version 10.3.1.

Results

Clinical and demographic information and statistical analysis of participant information are in Table 1. Thirteen female participants were included (ME/CFS: $n = 6$; healthy controls: $n = 7$). Mean age did not differ between groups (ME/CFS 48.2 ± 16.2 years; controls 45.5 ± 8.4 years; $p = 0.58$); age ranges were 29–65 and 35–57 years, respectively. Patients with ME/CFS had a mean Karnofsky Performance Score of 58.0 ± 11.0 and a MFI of 81.2 ± 5.1 (control values not collected). Hand grip strength was lower in the ME/CFS group (23.5 ± 10.4) compared with controls (32.6 ± 7.6), with a significant between-group difference ($p = 0.001$).

Table 1
Participant information.

Age	ME/CFS (n = 6)	Healthy control (n = 7)	p
	48.2 ± 16.2	45.5 ± 8.4	0.58
Age range	29–65	35–57	
Gender (M/F)	0/6	0/7	
Karnofsky score	58.0 ± 11.0	N/A	
Multidimensional Fatigue Inventory (MFI)	81.2 ± 5.1	N/A	
Hand Grip Strength	23.5 ± 2.1	32.6 ± 1.8	0.001
Note: data is presented as mean ± SD.			

iPSCs and MPs characterization

PBMCs were isolated and reprogrammed into iPSCs using episomal vectors expressing *Oct4*, *Sox2*, *Lin28*, *Klf4*, and *L-Myc*. All 13 lines exhibited typical iPSC morphology, forming tightly packed colonies with a high nucleus-to-cytoplasm ratio and prominent nucleoli (representative lines shown in Fig. 2A; images of all lines in Supplementary Fig. 1). Pluripotency was confirmed by immunofluorescence staining for TRA-1-60 and OCT4 (Fig. 2A; Supplementary Fig. 1), and by flow cytometry for SSEA4, EPCAM, and CD9 (Fig. 2D; Supplementary File 1). Trilineage differentiation potential was demonstrated following five days of directed differentiation, with immunofluorescence detection of ectoderm (PAX6, Nestin), endoderm (AFP, SOX17), and mesoderm (NCAM, Brachyury T) markers (Fig. 2B; Supplementary Fig. 2), supported by qPCR analysis (Fig. 2G).

All lines tested negative for mycoplasma by luminescence assay (Supplementary Table 1) and showed normal molecular karyotypes with genotypes identical to their parental PBMCs, as determined by SNP array (Supplementary File 2). No significant differences were observed between ME/CFS and control iPSCs in key pluripotency gene expression or proliferation capacity (Fig. 2E, 2F, 2G). Finally, upon myogenic differentiation, immunocytochemistry revealed expression of MyoD and MyoG, confirming specification toward the myogenic progenitor lineage (representative lines shown in Fig. 2C; full dataset in Supplementary Fig. 3).

RNA-Seq Data and differentially expressed genes

RNA sequencing was performed on 13 myogenic progenitor (MP) cell lines derived from iPSCs that passed quality control assessment. Principal component (PC) analysis showed that first two PCs captured 60% of the total variance in gene expression across the dataset, with PC1 and PC2 explaining 44% and 16% respectively (Supplementary File 3). Following normalisation, differential expression analysis identified 1,323 genes differentially expressed between ME/CFS and healthy controls at nominal $p < 0.05$ (Supplementary Table 3). After the false discovery rate (FDR) correction (adjusted $p < 0.05$), 7 genes remained significantly differentially expressed (Table 2). Detailed information on the known biological functions of these genes is presented in Table 2. Five genes (*MIR205HG*, *FEZF1*, *SLC1A2*, *SYT13*, and *GALNT4*) were selected for validation by qPCR. Of these, only *HLA-DMB* and *SYT13* were confirmed as significantly differentially expressed, with *HLA-DMB* downregulated ($p < 0.001$) and *SYT13* upregulated ($p = 0.004$) in ME/CFS compared with healthy controls (Table 2).

Table 2
DEGs with adjusted p < 0.05 and functions

Gene Symbol	Description	log2Fold Change	Adjusted p	log2Fold change (qPCR)	Biological function*
<i>MIR205HG</i>	MIR205 Host Gene (Non-Protein Coding)	-16.9	< 0.001	ND	<i>MIR205HG</i> is a long noncoding RNA that regulates prolactin production, involved in prolactin metabolic process, affects metabolic homeostasis by regulating key enzymes and transporters that are associated with glucose and lipid metabolism.
<i>FEZF1</i>	FEZ Family Zinc Finger 1	12.3	< 0.001	ND	<i>FEZF1</i> is a gene that encodes a transcriptional repressor and has been shown to repress the transcription factor HES5, which is activated downstream of the Notch pathway
<i>HLA-DMB</i>	Major Histocompatibility Complex, Class II, DM Beta	-5.7	0.008	-4.26 (p < 0.001)	<i>HLA-DMB</i> plays a critical role in antigen processing and presentation by facilitating peptide loading onto MHC class II molecules in antigen-presenting cells like dendritic cells and B cells.
<i>SLC1A2</i>	Solute Carrier Family 1 Member 2	-1.4	0.040	ND	<i>SLC1A2</i> encodes for the membrane-bound protein, a principal transporter that clears the excitatory neurotransmitter glutamate from the extracellular space at synapses in the central nervous system.
<i>SYT13</i>	Synaptotagmin 13	1.6	0.040	2.09 (p = 0.004)	<i>SYT13</i> belongs to a family of membrane-trafficking proteins, specifically those containing C2 domains of certain synaptotagmins that bind calcium. Synaptotagmins are proposed to function as calcium sensors involved in the regulation of neurotransmitter release and hormone secretion.
<i>GALNT4</i>	Polypeptide N-Acetylgalactosaminyltransferase 4	-0.8	0.047	NR	<i>GALNT4</i> gene encodes a member of the UDP-N-acetyl-α-D-galactosamine:polypeptide N-acetylgalactosaminyltransferase (GalNAc-T) family of enzymes. <i>GALNT4</i> initiates mucin-type O-linked glycosylation in the golgi apparatus by catalysing the transfer of GalNAc to serine and threonine residues on target proteins.
<i>SLC2A14</i>	Solute Carrier Family 2 Member 14	-2.4	0.047	NR	<i>SLC2A14</i> encodes a member of the solute carrier family 2 (facilitated glucose transporters), which is involved in glucose transport across cell membranes.

ND = Not detected; Ct value was undetermined or exceeded the threshold in all replicates, indicating no detectable expression under the experimental conditions using primers designed for known validated gene variants from NIH Entrez database. **NR** = Not run; gene not selected for qPCR validation. *Biological functions were extracted from GeneCards® - The Human Gene Database (www.genecards.org).

Pathway Enrichment Analysis

KEGG annotation analysis was employed to examine differential pathways in RNA-seq data from iPSC-MPs from both ME/CFS cases and healthy controls, aiming to identify distinct biological pathway signatures. A complete list of 358 enriched pathways is provided in Supplementary Table 4. Among these, 73 differentially expressed pathways with q-values < 0.05 were identified and are presented in Fig. 3A, the majority of which were downregulated (96%). Only the “autophagy” and “valine, leucine, and isoleucine degradation” pathways were upregulated (Fig. 3A). The top 20 pathways, revealed by both KEGG and GO analyses (Figs. 3B and 3C), showed substantial overlap, including two pathways associated with skeletal muscle structure: “cytoskeleton in muscle cells” and “regulation of actin cytoskeleton”. Most of the downregulated pathways were related to cell cycle and DNA maintenance, while a notable subset was associated with immune and inflammatory responses, and cell adhesion and cytoskeleton (Fig. 3D).

Drug repurposing candidates for treatment of ME/CFS

A subset of the top 100 upregulated and downregulated DEGs were queried against the LINCS2 database to identify potential drug candidates for repurposing. A complete list of identified compounds with an FDR < 0.05 was retained for further analysis (Supplementary Table 5). The top 100 compounds were then reviewed in detail, and drugs were selected based on a literature review, applying the pre-defined exclusion criteria. The final list of shortlisted repurposing candidates is presented in Table 3. There are 22 drugs that met the inclusion criteria and are currently approved for use in humans. Supporting evidence from the literature for each selected candidate is summarised in Supplementary Table 6.

Table 3
Selected drug candidates with repurposing potential for treatment of ME/CFS

Drug	NCS	WTCS - FDR	Mechanism of action	Approved therapeutic indication
Acyclovir	-1.55	0.003	DNA polymerase inhibitor	Cold sores Genital herpes
Tetrahydrobiopterin	-1.53	0.003	Nitric oxide synthase stimulant Phenylalanine 4-hydroxylase stimulant	Phenylketonuria
Leflunomide	-1.51	0.003	Dihydroorotate dehydrogenase inhibitor PDGFR receptor inhibitor	Active rheumatoid arthritis Multiple sclerosis
Naftidrofuryl	-1.50	0.003	Adrenergic receptor antagonist	Peripheral vascular disorders
Tacrine	-1.49	0.003	Acetylcholine release stimulant Acetylcholinesterase inhibitor Butyrylcholinesterase inhibitor Potassium channel antagonist	Alzheimer disease
Zolmitriptan	-1.48	0.003	Serotonin receptor agonist	Migraine with aura Migraine without aura
Nalbuphine	-1.46	0.003	Opioid receptor agonist Opioid receptor antagonist	Pain
Melatonin	-1.46	0.003	Serotonin receptor agonist	Insomnia, jet lag, circadian rhythm disorders
Erythromycin	-1.44	0.003	NFkB pathway inhibitor	Listeriosis Diphtheria Respiratory tract infections Bacterial endocarditis Staphylococcal dermatitis
Vinpocetine	-1.44	0.003	Phosphodiesterase inhibitor Sodium channel blocker	Vascular dementia
Sirolimus	-1.43	0.003	MTOR inhibitor	Lymphangiomyomatosis
Nitrendipine	-1.43	0.003	Calcium channel blocker	Hypertension
Noscapine	-1.42	0.003	Bradykinin receptor antagonist Tubulin inhibitor	Chronic Cough
Fenofibrate	-1.40	0.003	PPAR receptor agonist	Hypertriglyceridemia
Primidone	-1.40	0.003	GABA receptor antagonist	Focal epileptic seizures Psychomotor epileptic seizures

Drug	NCS	WTCS - FDR	Mechanism of action	Approved therapeutic indication
				Grand mal epileptic seizures
Everolimus	-1.39	0.003	MTOR inhibitor	HER2 negative breast cancer Neuroendocrine tumours Advanced renal cell carcinoma
Disulfiram	-1.37	0.003	Aldehyde dehydrogenase inhibitor DNA methyltransferase inhibitor TRPV agonist (especially post-infectious subtypes)	Chronic alcoholism
Clofibrilic-acid	-1.36	0.003	PPAR receptor agonist	Hyperlipoproteinemia
Perindopril	-1.36	0.003	ACE inhibitor	Stable coronary artery disease Hypertension
Zidovudine	-1.36	0.003	Reverse transcriptase inhibitor	HIV-1 infection
Donepezil	-1.35	0.003	Acetylcholinesterase inhibitor	Alzheimer disease
Midodrine	-1.35	0.003	Adrenergic receptor agonist	Orthostatic hypotension

NCS = Normalised Connectivity Score. This value represents the strength of the connection between a drug-induced gene expression signature and the ME/CFS gene expression profile. A more negative NCS indicates a greater potential for the drug to reverse the disease signature. The shortlisted drugs are ranked by NCS values.

WTCS-FDR = Weighted connectivity Score-False discovery rate. The WTCS-FDR represents the adjusted p-value (FDR) for a drug's connectivity score, estimating how likely the observed match/mismatch between a compound and a gene signature is due to chance.

Discussion

While numerous clinical trials have been conducted involving pharmacological compounds, nutraceuticals, and exercise therapy, no effective treatment has been approved for ME/CFS, primarily due to the limited understanding of its pathophysiology. Using iPSC-MPs, transcriptomic differences in cells derived from ME/CFS cases and healthy controls were identified in this study. Mapping the data using KEGG annotation revealed 73 significantly altered biological pathways. Subsequent analysis using the LINCS2 database identified 22 drugs with potential for repurposing to treat ME/CFS.

Differentially expressed genes between ME/CFS patients and healthy controls

This is the first study to identify transcriptomic signatures of ME/CFS using iPSC-MPs, with seven genes reaching significance based on an adjusted p-value < 0.05. Among these genes, some have previously been linked to ME/CFS in the literature. For example, *MIR205HG* is a non-long coding gene that regulates the prolactin expression, primarily through its modulatory effects on transcriptional and post-transcriptional signalling pathways [30]. Given the documented neuroendocrine dysregulation in ME/CFS, this finding adds a novel molecular link between altered gene expression and disrupted hormonal responses in affected individuals. Previous studies have reported heightened prolactin responses to serotonergic and dopaminergic challenge tests in ME/CFS patients, despite baseline prolactin levels being typically normal.

For instance, increased prolactin secretion was observed following d-fenfluramine and buspirone administration, suggesting hypersensitivity of serotonergic signalling pathways [31–33]. An exaggerated prolactin response to sulpiride is however described in depression, indexing possible dopamine supersensitivity [34]. The increase of prolactin hormonal responses have been proposed as potential biomarkers distinguishing ME/CFS from other fatigue-related conditions and may be associated with impaired hypothalamic-pituitary axis regulation [33]. It is possible that reduced *MIR205HG* expression diminishes its regulatory control over prolactin synthesis or release, contributing to the heightened sensitivity of the prolactin axis seen in ME/CFS. Furthermore, the sex-specific nature of prolactin regulation, particularly its stronger association in female patients [35] and the present study including only female samples, warrants additional investigation into whether *MIR205HG* downregulation exhibits sex-biased expression patterns in ME/CFS cohorts.

HLA-DMB (Major Histocompatibility Complex (MHC), Class II, DM Beta) was also identified and confirmed by qPCR, a gene that facilitates loading of antigenic peptides onto MHC class II molecules in antigen-presenting cells, particularly in dendritic cells and B cells, which play important roles in antigen processing and presentation [36]. Impaired expression of this gene can disrupt immune surveillance and contribute to immune dysregulation, leading to reduced efficiency in antigen recognition and an inadequate immune response [36, 37]. Several abnormalities in B cells have been reported in ME/CFS, including altered B cell phenotypes, reduced memory B cell numbers, and increased plasma blast differentiation [38–40]. While a recent transcriptomic study of PBMCs from ME/CFS patients reported upregulation of transcription factors associated with CD4⁺ and CD8⁺ T cell activation and inflammatory responses [41, 42], our data from iPSC-MPs identified a significant downregulation of *HLA-DMB*. Its reduced expression may impair local antigen presentation in skeletal muscle cells, thereby contributing to dysregulated CD4⁺ T cell activation or tolerance. Although direct evidence linking *HLA-DMB* to ME/CFS in the literature is limited, previous studies have reported associations between ME/CFS and other HLA class II alleles, including *HLA-DQA101** and *HLA-DQB103:03** [43]. Additionally, other HLA genes, particularly class II alleles, have also been implicated in post-infectious ME/CFS pathogenesis and altered responses to Epstein–Barr virus [44], a known environmental trigger of the disease. Our findings suggest that antigen presentation defects—specifically involving *HLA-DMB*—may extend beyond traditional immune cells, potentially contributing to aberrant T cell responses through impaired B cell–T cell crosstalk in muscle-resident progenitor compartments. This highlights the importance of examining non-haematopoietic cells in ME/CFS pathophysiology and supports a broader model of immune dysregulation involving both peripheral and tissue-specific components, warranting further investigation into its role in ME/CFS.

SYT13 is a member of the synaptotagmin family, and plays a role in vesicle trafficking, membrane repair, calcium-dependent exocytosis [45], and glucose transport to cells [46]. Its upregulation in genome-wide transcriptomics and qPCR observed in the present study could imply changes in intracellular calcium signalling [47] and intracellular glucose levels [46], which are critical in muscle contraction and mitochondrial energy regulation. Upregulation of *SYT13* may represent a compensatory response to cellular stress, such as impaired mitochondrial function or calcium dysregulation, both of which have been widely reported in ME/CFS and reviewed in recent publications [6, 48]. Additionally, *SYT13* has been implicated in regulating vesicle trafficking pathways that could influence intracellular glucose handling, including trafficking of glucose transporters (such as GLUTs) or metabolic enzymes to the cell membrane [2, 6]. Evidence from β -cell models has shown that reduced *SYT13* expression impairs insulin secretion by disrupting granule trafficking, highlighting its importance in maintaining secretory and metabolic function [6]. Although skeletal muscle is not a classical secretory tissue, upregulation of *SYT13* may reflect an adaptive mechanism to preserve intracellular glucose uptake or utilisation under energy-restricted conditions. Notably, mitochondrial dysfunction and impaired glycolytic flux have been observed in ME/CFS skeletal muscle and immune cells, consistent with cellular stress responses that might drive *SYT13* expression [3, 7, 8].

Significantly enriched pathways

KEGG and GO pathway analysis revealed most significant differential pathways were downregulated in ME/CFS compared with healthy controls. These pathways were mainly linked to cell cycle and DNA repair pathways, immune-related pathways, and pathways related to muscle function, cytoskeleton in muscle cells and regulation of the actin cytoskeleton pathways.

Cell cycle and DNA repair pathways

The observed downregulation of cell cycle and DNA repair pathways including p53 signalling, mismatch repair, and base excision repair in ME/CFS-derived MPs suggests a broad impairment in their proliferative and regenerative potential. These pathways are essential for maintaining genomic integrity, facilitating proper mitotic progression, and executing damage response mechanisms during cellular stress [49–52]. Suppression of the p53 signalling pathway, in particular, may reflect an imbalance between cell cycle arrest, apoptosis and DNA repair activation as cells respond to stress which can drive accumulated DNA damage and cellular senescence [49, 53]. Mondarano *et al.* (2020) reported significant telomere shortening in ME/CFS patients, especially in CD8 + T cells and NK cells, suggesting replicative exhaustion and reduced regenerative potential of immune cells [54]. Telomere attrition, which activates DNA damage responses and p53 signalling, may persist due to inefficient resolution of genotoxic stress. Additionally, mitochondrial dysfunction, widely reported in ME/CFS [6, 7, 48], may further exacerbate genomic instability. Reactive oxygen species generated by dysfunctional mitochondria can induce oxidative DNA damage, necessitating robust repair via base excision and mismatch repair pathways [9]. Impairment of these systems in MPs could therefore reflect both an intrinsic vulnerability and a secondary consequence of chronic metabolic stress.

Immune-related pathways

Evidence of immune dysfunction in ME/CFS is well documented, with numerous studies reporting the impairments of genes/pathways related to the immune response in affected individuals [38–42, 44, 55, 56]. Our KEGG-based transcriptomic analysis of iPSC-derived muscle progenitors (MPs) revealed broad enrichment in immune-related pathways, with 26 out of 73 significantly altered pathways associated with immune function. In contrast to transcriptomic studies using peripheral immune cells, which report widespread upregulation of immune and viral response pathways in ME/CFS patients [38, 40, 41], our analysis showed broad downregulation of these pathways in ME/CFS-derived MPs.

This discrepancy may reflect tissue-specific immune regulation. In peripheral blood mononuclear cells (PBMCs), chronic activation of monocytes and T cells as evidenced from sustained expression of inflammatory mediators, has been linked to immune cell exhaustion, characterised by diminished responsiveness and altered transcriptional profiles [38, 40, 41, 55]. This exhausted state may impair these immune cells' ability to coordinate systemic immune responses, including communication via cytokines and chemokines with peripheral tissues such as muscle. Our study observed a downregulation of multiple key immune signalling pathways, such as cytokine-cytokine receptor interaction, TNF signalling, IL-17 signalling, JAK-STAT signalling, and T-cell receptor signalling. These pathways play essential roles in mediating immune communication, inflammation, and cellular responses to infection or injury [57, 58]. Their suppression suggests a diminished capacity of muscle progenitor cells to respond to systemic immune cues or inflammatory signals. In particular, the cytokine-cytokine receptor interaction pathway integrates upstream signalling from inflammatory mediators [59], while the JAK-STAT and T-cell receptor pathways are critical for downstream gene transcription and T cell activation [57]. The TNF and IL-17 signalling cascades further coordinate inflammatory and immune cell recruitment processes [58, 60]. Therefore, the broad suppression of these signalling modules in MPs may indicate impaired immune responsiveness within skeletal muscle, which may limit regenerative capacity and underlie the impaired recovery and post-exertional symptom exacerbation characteristic of ME/CFS. In addition to canonical immune signalling suppression, our KEGG analysis also revealed downregulation of several viral-related immune pathways, including those associated with Epstein-Barr virus infection, cytomegalovirus infection, Kaposi sarcoma-associated herpesvirus infection, and viral protein-cytokine interaction. These pathways typically reflect the host cell's capacity to detect and respond to latent or active viral infections through cytokine signalling, interferon responses, and antigen processing [61]. Their downregulation in ME/CFS-derived MPs may reflect impaired viral recognition and clearance in muscle tissue, contributing to persistent immune dysregulation. Taken together, while circulating immune cells in ME/CFS patients may exhibit chronic activation, MPs display a distinct pattern of blunted immune responsiveness. This cell-type specific immune modulation supports a model in which ME/CFS pathophysiology involves differential immune states across tissues, with systemic immune activation coexisting alongside local immune suppression or signalling failure in affected peripheral tissues such as skeletal muscle.

Cytoskeleton pathways

There are several differentially expressed pathways related to muscle function such as cytoskeleton in muscle cells and regulation of the actin cytoskeleton pathways. The cytoskeleton in muscle cells pathway involves a network of structural and regulatory proteins essential for maintaining muscle cell integrity, facilitating contraction, and mediating mechanotransduction. Key components include actin and myosin filaments, sarcomeric structures, costameres, desmosomes, intermediate filaments such as desmin, and the LINC complex, which links the nucleus to the cytoskeleton. These components work together to preserve muscle cellular structure and transmit mechanical force during muscle contraction [62, 63]. Disruptions in this pathway have been linked to various skeletal muscle disorders, including Duchenne muscular dystrophy [64], desminopathies [65], laminopathies [66], and congenital myopathies [67]. In the context of ME/CFS, dysregulation of genes within this pathway may impair muscle structure and contractile function, contributing to the hallmark symptoms of fatigue and muscle weakness [5, 6]. Our findings reveal that several genes involved in the “Cytoskeleton in Muscle Cells” pathway were significantly downregulated in iPSC-derived MP cells from ME/CFS patients, including *COL9A1*, *ITGA11*, *ITGA2*, *PKP2*, *COL9A3*, *SDC1*, *DAAM2*, *DIAPH3*, and *FBN3*. These genes play critical roles in extracellular matrix organisation, cell adhesion, and actin cytoskeleton regulation, suggesting that their altered expression may contribute to cytoskeletal instability and reduced mechanical efficiency in ME/CFS muscle cells.

The regulation of the actin cytoskeleton pathway plays a central role in maintaining cellular architecture, intracellular trafficking, signal transduction, and contractile force generation, particularly in muscle cells. Actin filaments are dynamic structures regulated by a range of signalling molecules such as Rho GTPases, kinases (e.g., ROCK, PAK), and actin-binding proteins, which coordinate polymerisation and depolymerisation processes to adapt to mechanical and metabolic demands [68]. In skeletal muscle, correct organisation and regulation of actin filaments is essential for sarcomere structure, muscle contraction, and repair following stress or injury [62]. Disruption of this pathway can lead to altered muscle fibre structure, impaired contractility, and abnormal cell signalling. These abnormalities may contribute to muscle fatigue and reduced physical endurance, which are hallmark symptoms of ME/CFS. Previous studies have suggested that ME/CFS is associated with impaired energy metabolism and muscle dysfunction, potentially linked to abnormalities in cytoskeletal regulation [55, 69]. Altered expression of genes involved in actin cytoskeleton regulation may therefore provide insight into the cellular mechanisms underlying muscle symptoms in ME/CFS and serve as potential targets for therapeutic intervention. Findings from this study show that the regulation of genes involved in the actin cytoskeleton pathway is downregulated in iPSC-derived MP cells from ME/CFS patients, highlighting the need for further investigation to evaluate their potential as disease biomarkers or therapeutic targets.

Upregulation of amino acids degradation pathway

The upregulation of the valine, leucine, and isoleucine degradation pathway in ME/CFS-derived MPs suggests a metabolic shift toward increased reliance on branched-chain amino acids (BCAAs) for energy production. BCAAs are primarily metabolised in peripheral tissues such as skeletal muscle and serve as key alternative fuel sources, feeding intermediates like acetyl-CoA and succinyl-CoA into the tricarboxylic acid (TCA) cycle, especially under conditions of glucose deprivation or mitochondrial stress [70, 71]. This upregulation may reflect a compensatory adaptation to the energetic challenges frequently reported in ME/CFS, including mitochondrial dysfunction, reduced glycolytic flux, and impaired fatty acid oxidation [72, 73]. Several metabolomic and transcriptomic studies in ME/CFS cohorts have reported altered BCAA levels, with some suggesting increased BCAA catabolism to support ATP production in the face of systemic energy deficits [74, 75]. Increased degradation of BCAAs may also attempt to counteract the accumulation of toxic intermediates or oxidative stress. For example, while moderate BCAA breakdown replenishes TCA intermediates, overactivation may lead to excess production of branched-chain keto acids (BCKAs) and reactive species, further exacerbating mitochondrial strain [74, 75]. Thus, while the transcriptional upregulation of BCAA degradation may reflect a metabolically adaptive response, it may also signal a state of metabolic inflexibility in ME/CFS muscle progenitors. Furthermore, BCAA catabolism intersects with important metabolic signalling pathways such as mTOR, AMPK, and PPAR α , which regulate autophagy, protein synthesis,

and mitochondrial biogenesis [76, 77]. Increased flux through BCAA degradation could therefore influence cell growth, differentiation, and stress responses in MPs. However, the long-term effects of sustained BCAA catabolism are not necessarily beneficial, particularly if it reflects chronic compensation for deficits in primary metabolic pathways.

Interestingly, the upregulation of this pathway in MPs aligns with broader hypotheses in ME/CFS that propose muscle and immune cells are forced to shift to less efficient energy production pathways due to underlying mitochondrial impairments [41, 78–80]. In this context, increased BCAA degradation may serve to buffer TCA cycle activity, sustain ATP production, and mitigate energetic collapse. These results add to a growing body of evidence that ME/CFS involves not only immune and mitochondrial dysfunction, but also complex metabolic reprogramming. Whether this shift represents a protective adaptation or a contributor to metabolic exhaustion remains to be clarified, but it clearly underscores the bioenergetic strain faced by MPs in ME/CFS.

Potential drug candidates for repurposing ME/CFS

Using transcriptomic profiles derived from ME/CFS MPs and LINCS2-based reverse gene signature screening, we identified a shortlist of candidate drugs with potential to reverse ME/CFS-associated gene expression patterns. Following literature-based screening, 22 compounds are proposed for further investigation, representing diverse biological mechanisms relevant to ME/CFS pathology. A notable subset of these drugs includes antiviral or antimicrobial agents, such as acyclovir, zidovudine, erythromycin, and disulfiram. Their identification is consistent with extensive literature implicating persistent or latent viral infections, particularly Epstein-Barr virus and human herpesvirus 6, in ME/CFS onset and progression [2, 6, 39, 41–43, 55]. Disulfiram, an inhibitor of viral proteases, for instance, has shown activity against intracellular pathogens and has recently gained attention for its potential to reactivate and clear latent viruses such as HIV [81], and reduce SARS-CoV-2 infection [82]. These drugs may support viral clearance and help resolve chronic immune activation observed in ME/CFS patients.

Several candidates also fall under the category of immunomodulators, including leflunomide, sirolimus, everolimus, clofibric acid, and fenofibrate. These agents modulate key immune pathways implicated in ME/CFS, such as T-cell activation, cytokine regulation, and mTOR signalling [83–86]. Both sirolimus and everolimus inhibit mTORC1 [87], a metabolic and immune checkpoint regulator that has been associated with immune exhaustion and metabolic reprogramming in T cells [88]. The identification of PPAR- α agonists (e.g. fenofibrate, clofibric acid) is also relevant, as they modulate inflammation, lipid metabolism, and regulate mitochondrial function [89], all of which have been implicated in the pathophysiology of ME/CFS [83–86].

Drugs with neuroprotective or neurotransmitter-modulating effects also featured prominently including tacrine, donepezil, and tetrahydrobiopterin (BH4). These drugs may impact cognitive function and address neurotransmitter dysregulation [90, 91], which correspond with observed symptoms in ME/CFS such as cognitive fog and central fatigue. BH4, a cofactor for monoamine synthesis, could help restore serotonergic and dopaminergic tone [91], both implicated in ME/CFS pathophysiology and hypothalamic-pituitary axis dysfunction. Additionally, melatonin, a serotonin receptor agonist may have multifaceted effects by regulating circadian rhythm, sleep patterns, neuroprotective effects, mitochondrial oxidative stress, and immune responses [92, 93]. Notably, clinical trials reported that ME/CFS patients showed improvements in fatigue, motivation, concentration and activity after treatment with 1-5mg of melatonin for three months [92, 94]. Thus, these selected neuroprotective drugs have potential for ME/CFS repurposing; however, further studies are needed to holistically evaluate their treatment doses, times and ME/CFS subtype efficacy.

Drugs targeting autonomic dysfunction and vascular dysregulation, including midodrine, naftidrofuryl, perindopril, and nitrendipine, may alleviate orthostatic intolerance and improve cerebral perfusion, common symptoms in ME/CFS. Midodrine for instance, which is an anti-hypotensive compound that increases blood pressure by activating the alpha 1-receptor, has already been used to treat postural orthostatic tachycardia syndrome (POTS), which frequently overlaps with ME/CFS [95]. In a case study, Naschitz *et al.* (2004) reported that ME/CFS patients with abnormal haemodynamic instability

score treated with midodrine resulted in correction of the dysautonomia followed by improvement of fatigue [96]. They suggested that midodrine might be used for ME/CFS patients with autonomic dysfunction; however, there are no further placebo-controlled trials on the efficacy of midodrine on ME/CFS since.

The short list of potential drugs also includes zolmitriptan, nalbuphine, and noscapine. These drugs may offer symptomatic relief for headaches, pain, or the neuroinflammatory aspects of the disease. While these agents may not directly modify disease mechanisms, they may contribute to multi-modal treatment strategies.

Limitations and future directions

Despite the strengths of this study, several limitations should be acknowledged. First, the sample size was limited, and although consistent transcriptomic differences were observed, larger cohorts may be required to validate the generalisability of these findings across the heterogeneous ME/CFS population. For example, all participants in this study were female, limiting the applicability of our findings to male patients, who may exhibit distinct molecular signatures. While iPSC-MPs provide a muscle-relevant platform that captures patient-specific molecular alterations, they do not fully recapitulate the complex multisystemic features of ME/CFS, including immune-neuroendocrine interactions and post-exertional symptom exacerbation. Further, *in vitro* and *in vivo* functional validation of identified compounds remains necessary to confirm therapeutic potential and disease relevance. Future studies should aim to incorporate multi-tissue iPSC models, longitudinal sampling, and single cell analyses to better capture disease dynamics and cellular heterogeneity. Integration with proteomic, metabolomic, and clinical phenotyping datasets may further advance biomarker discovery and precision medicine approaches for ME/CFS.

We present the first transcriptomic characterisation of patient-derived iPSC-MPs in ME/CFS, establishing a novel disease-relevant *in vitro* model to investigate pathogenic mechanisms and therapeutic opportunities. The identified drugs span key pathophysiological domains identified in ME/CFS - immune dysregulation, neuroinflammation, autonomic instability, mitochondrial impairment, and viral persistence, underscoring the potential of patient-specific iPSC-MPs to serve as a scalable and mechanistically informative platform for ME/CFS modelling and rational drug repurposing. The present study adds to the growing utility of GES of diseases that are then used for screening drug candidates [97, 98] such as candesartan and trimetazidine to treat bipolar depression, which are currently undergoing clinical trials. By leveraging a model that captures disease-relevant molecular features and aligns with known pharmacological mechanisms, this approach can address key barriers in ME/CFS research, including its elusive aetiology, lack of validated biomarkers, and the current absence of effective therapeutics.

Declarations

Ethics approval and consent to participate

Ethics approval for this study was obtained from the Barwon Health Human Research Ethics Committee (20/202) and the Deakin University Human Research Ethics Committee (2021 – 165), Australia. The study was conducted in accordance with the ethical requirements approved by these Human Research Ethics Committees and the Australian National Statement on Ethical Conduct in Human Research. Written informed consent was obtained from all participants before participation.

Consent for publication

Not applicable.

Competing interests

The authors have no competing interests to declare.

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Ken Walder (ken.walder@deakin.edu.au).

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Author Contributions

Conceptualisation: KW, MB, CB, JHK, BP, DN

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Data and code availability

RNA-seq data, original code, and any additional information required to reanalyse the data reported in this paper will be made available upon manuscript acceptance and can also be obtained from the lead contact upon reasonable request.

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Figures

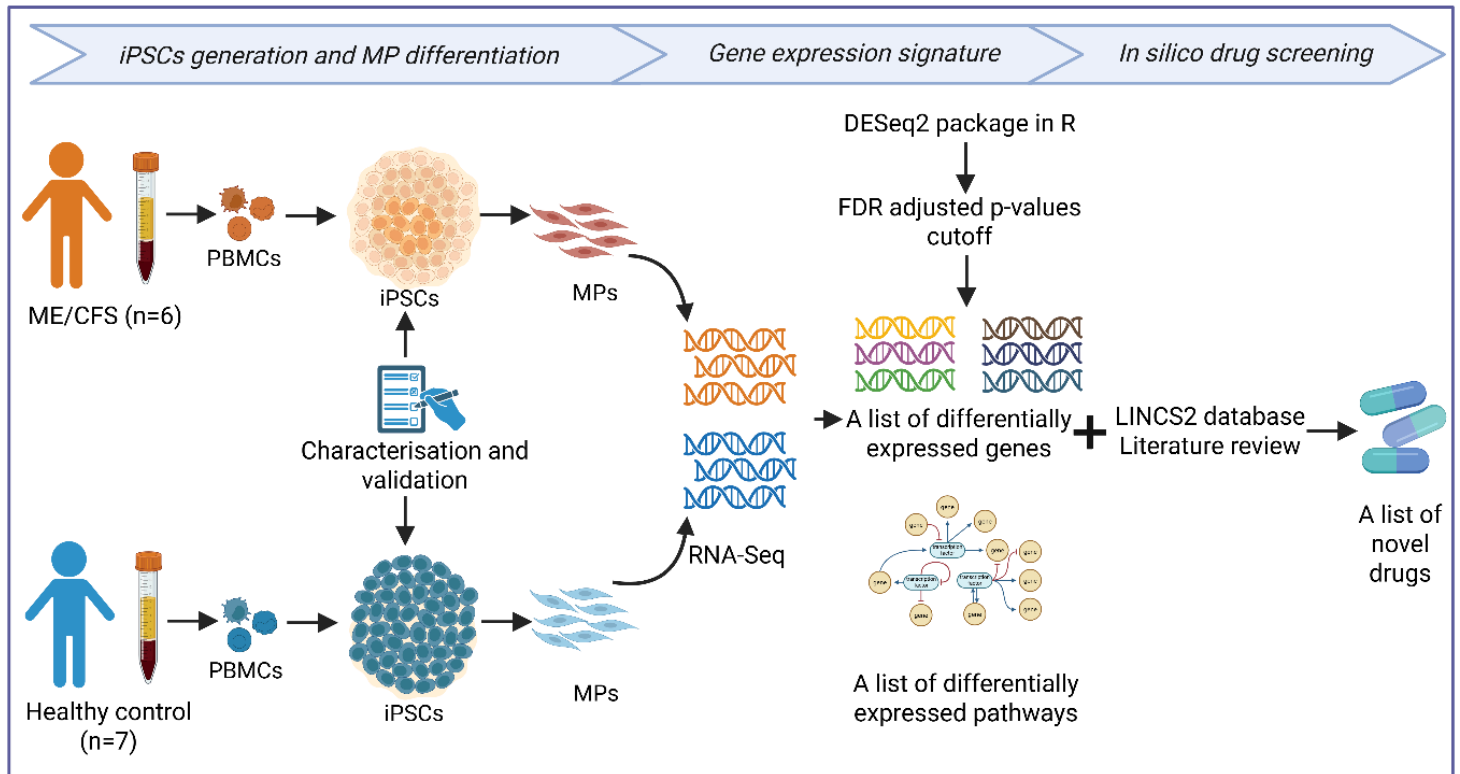


Figure 1

Study workflow from iPSC generation to in-silico drug prioritisation. Blood samples were collected from both ME/CFS patients and healthy controls. Induced pluripotent stem cells (iPSCs) were generated from peripheral blood mononuclear cells (PBMCs) and intensively characterised to confirm their pluripotency. Myogenic progenitor cells (MPs) were then differentiated from the iPSCs and used for RNA sequencing, differential expression and pathway enrichment analyses, and *in silico* drug screening.

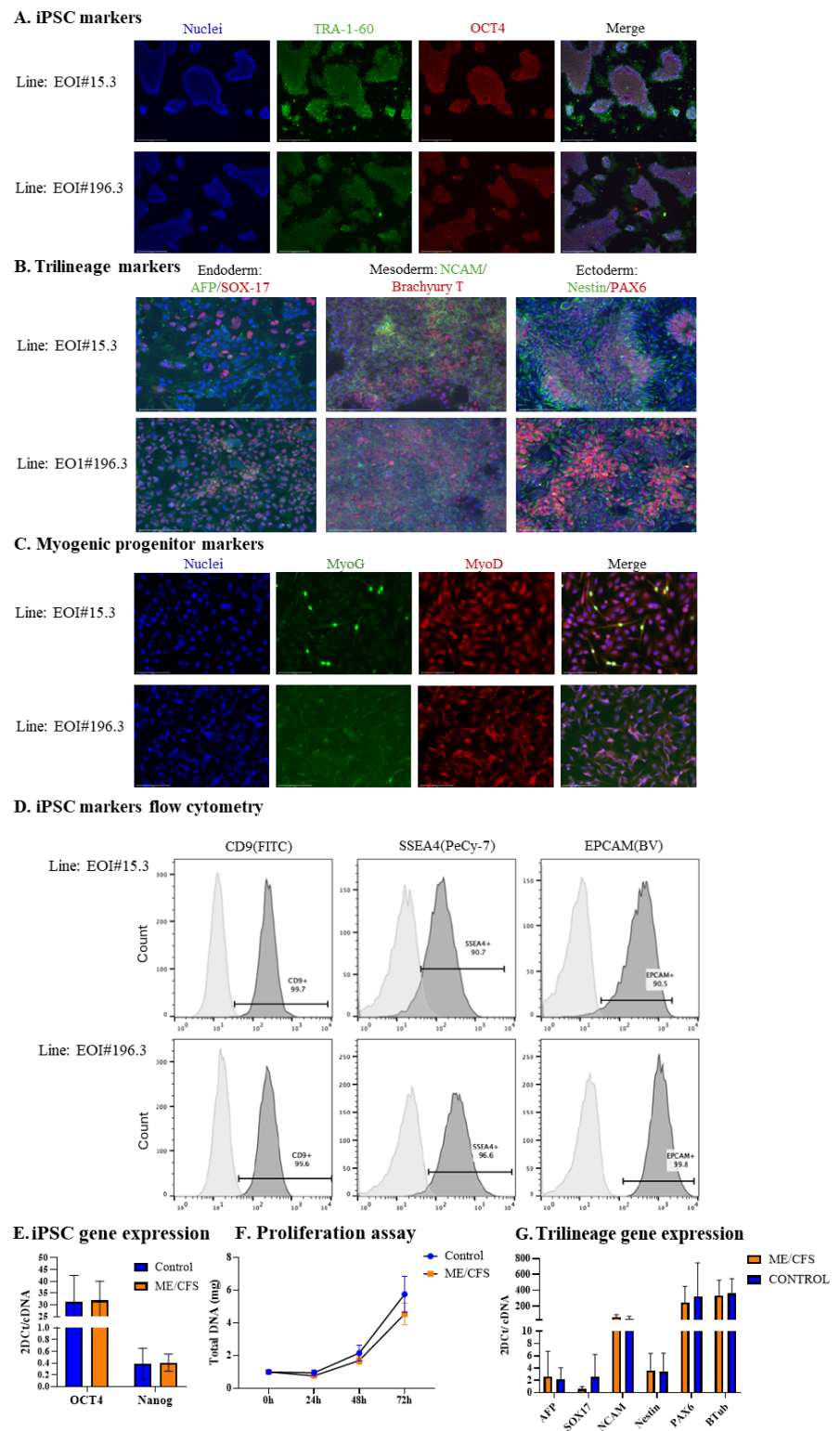


Figure 2

iPSCs and MPs characterisation. All 13 iPSC lines exhibited typical morphology (Figure 2A) and proliferation capacity (Figure 2F), expressed pluripotency markers by immunofluorescence (TRA-1-60, OCT4; Figure 2A, scale bar = 275 μ m) and flow cytometry (CD9, SSEA4 and EPCAM; Figure 2D), demonstrated differentiation potential into ectoderm, mesoderm, and endoderm lineages by immunofluorescence (Figure 2B, scale bar = 150 μ m) and qPCR (Figure 2G), and exhibited MyoD and MyoG progenitor markers (Figure 2C, scale bar = 150 μ m). No significant differences were observed between ME/CFS and control lines in key gene expression or proliferation (Figure 2E, 2G). Additional information for all lines in supplementary

materials. Panels A, B, C and D are representative cell lines; full data for all cell lines are in Supplementary Files 1 and Supplementary Figure 1, 2 and 3.

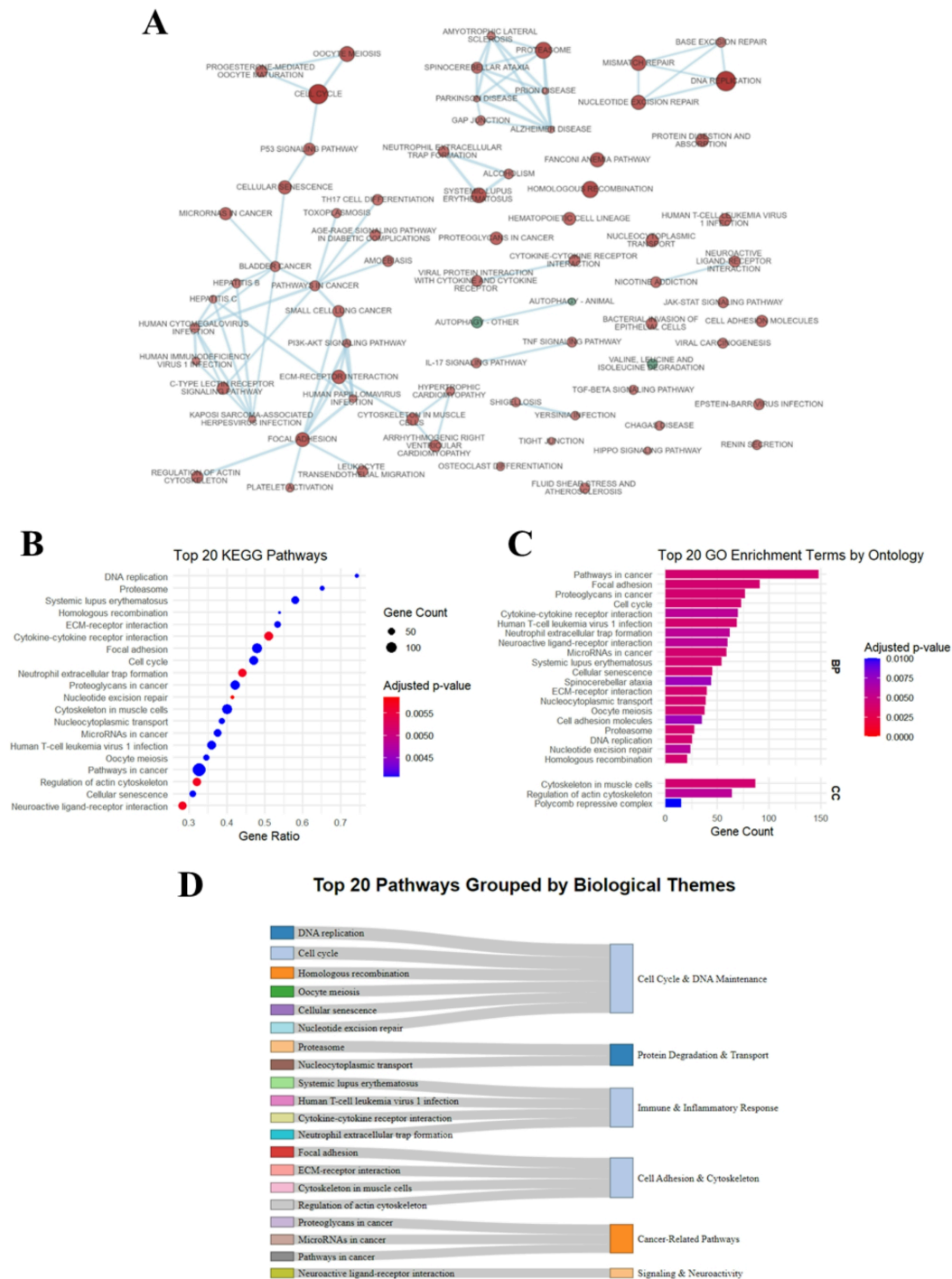


Figure 3

GSEA results using KEGG and GO pathways (**A**). Enrichment map of GSEA results in MP using KEGG pathway analysis, each node represents a significantly enriched pathway ($q\text{-value} < 0.05$), node size reflects the absolute Normalized Enrichment Score (NES); **green** indicates positive NES (upregulated), and **red** indicates negative NES (downregulated), edge thickness corresponds to the number of overlapping genes between pathways. (**B**) top 20 differential pathways of ME/CFS

vs healthy controls using KEGG pathways analysis, **(C)** top 20 differential pathways of ME/CFS vs healthy controls using GO ontology analysis. **(D)** top 20 KEGG pathways grouped by biological theme. BP=Biological processes, CC=cellular components.